Software Requirements Specification (SRS) System Name : MacroSight™ – Macroeconomic-Driven Sales Intelligence Platform

Document Type: Commercial Software Requirements Specification

1. Introduction

This document specifies the functional and non-functional requirements for a web-based Economic-Aware Sales Forecasting Platform. The system supports commercial organizations in forecasting sales while accounting for macroeconomic factors, tax policy changes issued by revenue authorities (e.g., IRS), and government legislation

. 1.1 Purpose The purpose of this SRS is to provide a clear and unambiguous description of system behavior, constraints, and interfaces for stakeholders, developers, and clients.

1.2 Scope The system ingests historical sales data and external economic indicators to generate forecasts, scenario simulations, and executive decision-support insights.

2. Overall Description

2.1 ***Product Perspective*** The platform is a standalone web-based application built on a service-oriented architecture, integrating analytics services and visualization dashboards.

2.***2 User Classes***

1 Business Executives – strategic planning and scenario analysis

2 Data Analysts – forecast evaluation and reporting

3 **System Administrators** – system configuration and access control

2.3 ***Operating Environment*** The system operates on modern web browsers and cloud-hosted Linux servers using containerized services.

3. Functional Requirements

3.1 ***Sales Data Management*** The system shall allow users to upload, validate, store, and manage historical sales data.

3.***2 Forecast Generation*** The system shall generate sales forecasts using statistical and machine learning models with configurable horizons and confidence intervals.

3.3 Macroeconomic Integration The system shall integrate inflation, exchange rate, and fuel price indicators as exogenous variables.

3.4 Tax Policy (IRS) Event Modeling The system shall register and encode tax policy changes issued by revenue authorities and incorporate them into forecasting and scenario simulations.

3.5 Government Legislation Analysis The system shall model non-numeric legislation through policy tagging and impact scoring to assess regulatory influence on sales.performance.

3.6 Scenario Simulation The system shall allow users to simulate hypothetical economic or policy changes and compare forecast outcomes side by side.

4. External Interface Requirements

4.1 User Interface The user interface shall be web-based, responsive, role-based, and provide charts, tables, and annotated timelines.

4.2 API Interfaces The backend shall expose RESTful APIs for data ingestion, forecasting, and reporting.

5. Non-Functional Requirements Performance:

Forecast requests shall complete within defined business thresholds.

Security: Role-based access control and secure authentication shall be enforced.

Scalability: The system shall support incremental data and user growth.

Reliability: Forecast outputs shall be reproducible and auditable.

6. Assumptions and Constraints The system assumes availability of reliable economic and policy data sources. The platform provides analytical insights and does not perform tax filing, payment, or legal compliance functions.

7. Future Enhancements Future versions may include automated data feeds, advanced machine learning models, and integration with enterprise planning systems.